

A. complex no. z is an ordered pair (x, y) of real nos x & y .
 $i = (0, 1)$

$$x = \operatorname{Re}(z) \quad \& \quad y = \operatorname{Im}(z).$$

$$z_1 = z_2 \Leftrightarrow \operatorname{Re} z_1 = \operatorname{Re} z_2 \quad \& \quad \operatorname{Im} z_1 = \operatorname{Im} z_2.$$

Sum: $z_1 + z_2 = (x_1 + x_2, y_1 + y_2)$ Add componentswise
 (commutes)

Product: $z_1 z_2 = (x_1 x_2 - y_1 y_2, x_2 y_1 + x_1 y_2) = z_2 z_1$
 (commutes)

In particular, if $y_1 = y_2 = 0$, then we get

$$(x_1, 0) + (x_2, 0) = (x_1 + x_2, 0)$$

$$(x_1, 0) \times (x_2, 0) = (x_1 x_2, 0)$$

B. these are rightly the extensions of real nos.

~~$z = x + iy$~~ For convenience, we write $z = x + iy$ which
 is nothing but $(x, 0) + (0, 1)(y, 0)$.

$$\text{as } (0, 1)(y, 0) = (0, y)$$

Thus if we denote i as $(0, 1)$, we can rewrite $z = x + iy$.

(Note: ~~where~~ $(0, 1)(0, 1) = (-1, 0) = -1$)

Associative laws: (i) $(z_1 + z_2) + z_3 = z_1 + (z_2 + z_3)$

(ii) $(z_1 z_2) z_3 = z_1 (z_2 z_3)$.

Thus, $z = x + iy = x + yi$

Additive identity: $(0, 0) = 0$.

Inverse: $-z$

Multiplicative identity: $z \cdot 1 = 1 \cdot z = z$.

Inverse: $\frac{1}{z}$ (for $z \neq 0$)

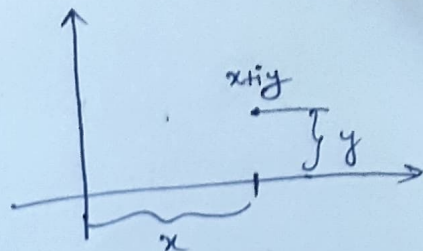
Now, subtraction is also defined as $z_1 + (-z_2)$.

division

$$z_1 \times \frac{1}{z_2} = z_1 z_2^{-1} \quad (z_2 \neq 0)$$

Complex plane (Argand Plane)

$z = x + iy$ or (x, y) is



Complex conjugate of $z = x + iy$ is denoted as $\bar{z}$

& defined as $\bar{z} = x - iy$

It is the reflection of z wrt Real axis.

Properties :-

$$z\bar{z} = x^2 + y^2$$

$$z + \bar{z} = 2x = 2 \operatorname{Re} z$$

$$z - \bar{z} = 2iy = 2i \operatorname{Im} z$$

$$\overline{z_1 \pm z_2} = \bar{z}_1 \pm \bar{z}_2$$

$$\overline{z_1 z_2} = \bar{z}_1 \bar{z}_2, \quad \overline{\left(\frac{z_1}{z_2}\right)} = \frac{\bar{z}_1}{\bar{z}_2}$$

~~Arg~~ Complex Conjugation commutes wrt $+$, $-$, $\times$, $\div$.

Ex. 1) $z = x + iy$ is ~~also~~ purely imaginary

$$\Leftrightarrow \bar{z} = -z$$

Polar co-ordinates

Instead of writing complex nos z in terms of xy co-ordinates, we also employ polar co-ordinates

$$x = r \cos \theta$$

$$y = r \sin \theta$$

$$z = r(\cos \theta + i \sin \theta)$$

r - absolute value or modulus of z , denoted $|z|$.

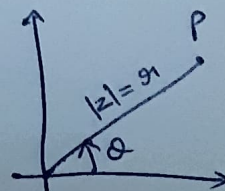
$$|z| = r = \sqrt{x^2 + y^2} = \sqrt{z\bar{z}}$$

θ - argument of $z = \arg z$ & $\tan \theta = \frac{y}{x}$ if $z \neq 0$

directed angle from the x -axis to OP

i.e. in the counterclockwise sense

(θ - measured in radians)



θ is undefined if $z=0$.

(3)

Since θ Let $z \neq 0$. And $\arg(z)$ is determined only upto integer multiples of 2π as $\sin$ & $\cos$ are 2π periodic. So, we define

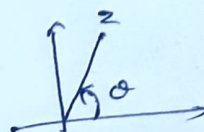
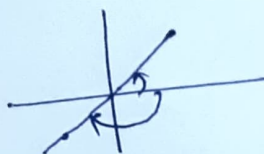
Principal value of argument of z ~~Arg z~~ ^{is} that a unique value

$\text{Arg } z$ such ~~Arg z~~ that $-\pi < \text{Arg } z \leq \pi$. Thus,

$$\arg z = \text{Arg } z + 2n\pi, \quad n \in \mathbb{Z}$$

Note:- If z is a $\neq$ real no., $\text{Arg } z = \pi$ (& not $-\pi$)

Eg $\text{Arg}(1+i) = \frac{\pi}{4}$, but $\text{Arg}(-1-i) = -\frac{3\pi}{4}$, & not $\frac{5\pi}{4}$



Euler's formula

$e^{i\theta}$ or $\exp(i\theta) = \cos \theta + i \sin \theta$ are points on the unit circle $|z|=1$.

Exponential form: $z = r e^{i\theta}$

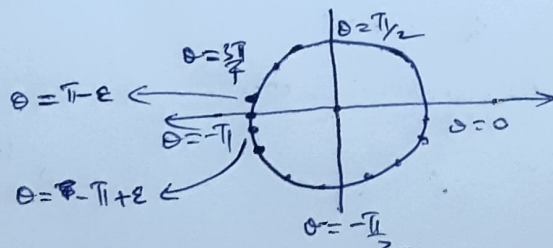
Let $z_1 = r_1 \cos \theta_1 + i r_1 \sin \theta_1$ & $z_2 = r_2 \cos \theta_2 + i r_2 \sin \theta_2$ $z_1 = r_1 e^{i\theta_1}$ & $z_2 = r_2 e^{i\theta_2}$

Multiplication

$$z_1 z_2 = r_1 r_2 e^{i(\theta_1 + \theta_2)}$$

The equation $z = r e^{i\theta}$ $-\pi < \theta \leq \pi$ is the parametric representation of the circle $|z|=r$, centered at origin with radius $r>0$.

As θ increases from $-\pi$ to π , the point z starts from the $-r$ real axis & traverses the circle once in the CCW direction.



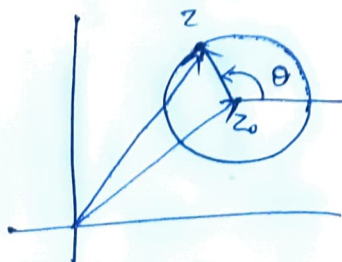
$$\theta \in (-\pi, \pi] \xrightarrow{\quad} \phi$$

$$0 \longmapsto r e^{i\theta}$$

More generally, any pt z on the circle centered at z_0 with radius r (4) has the parametric representation

$$z = z_0 + r e^{i\theta}, \quad -\pi < \theta \leq \pi.$$

$$\text{or } |z - z_0| = r$$



Eg 1) $e^{i\pi} = -1$

$$e^{2\pi i} = 1 = e^{-4\pi i}$$

$$e^{-i\pi/2} = -i$$

Exercises

1) $e^{i\theta_1} \cdot e^{i\theta_2} = e^{i(\theta_1 + \theta_2)}$

$$\begin{aligned} \text{LHS} &= (\cos \theta_1 + i \sin \theta_1)(\cos \theta_2 + i \sin \theta_2) = \cos \theta_1 \cos \theta_2 - \sin \theta_1 \sin \theta_2 \\ &\quad + i(\sin \theta_1 \cos \theta_2 + \cos \theta_1 \sin \theta_2) \end{aligned}$$

$$= \cos(\theta_1 + \theta_2) + i \sin(\theta_1 + \theta_2)$$

$$= e^{i(\theta_1 + \theta_2)}.$$

2) ~~$z = r e^{i\theta} \Rightarrow z^{-1} = \frac{1}{r} e^{-i\theta}$~~ More generally,
 $(r_1 e^{i\theta_1}) \cdot (r_2 e^{i\theta_2}) = r_1 r_2 e^{i(\theta_1 + \theta_2)}.$

2) If $z_1 = r_1 e^{i\theta_1}$ & $z_2 = r_2 e^{i\theta_2}$, then
 $(z_2 \neq 0)$

$$\frac{z_1}{z_2} = \frac{r_1}{r_2} e^{i(\theta_1 - \theta_2)} \Rightarrow \frac{1}{z} = \frac{1}{r} e^{-i\theta}.$$

3) $z^n = r^n e^{in\theta}$ for $n \geq 1$

Prove using mathematical induction.

Further $z^n = r^n e^{in\theta}$ for $n \leq 0$ by applying the above on $\frac{1}{z}$ ($z \neq 0$).
 (if $z \neq 0$)

Ex 4) Show that

$$(\cos \theta + i \sin \theta)^n = \cos n\theta + i \sin n\theta \quad \forall n \in \mathbb{Z}$$

Use 3 with $r=1$ or use induction to prove independently.

lec 14: Powers & roots

①

Let ~~z~~ Recall that $z = x+iy$ (in terms of x & y co-ordinates)
 $(z \neq 0)$
 $= r(\cos \theta + i \sin \theta)$ (in terms of polar co-ordinates)
 $= re^{i\theta}$ (in terms of exponential form)

where $r = |z|$ & θ is any real no. that satisfies $\tan \theta = \frac{y}{x}$ if $x \neq 0$

~~& if $x=0$, $\theta = \begin{cases} \frac{\pi}{2} + 2n\pi & \text{if } y > 0 \\ -\frac{\pi}{2} + 2n\pi & \text{if } y < 0 \end{cases}$~~

& if $x=0$,

~~$\theta = \begin{cases} \frac{\pi}{2} + 2n\pi \\ -\frac{\pi}{2} + 2n\pi \end{cases}$~~

θ is of the form

$\frac{\pi}{2} + 2n\pi$ if $y > 0$

$-\frac{\pi}{2} + 2n\pi$ if $y < 0$.

* In terms of z , $r = |z|$ & $\theta \in \arg z$.

Now,

$$z_1 z_2 = r_1 e^{i\theta_1} r_2 e^{i\theta_2} = r_1 r_2 e^{i(\theta_1 + \theta_2)}$$

$$\frac{z_1}{z_2} = \frac{r_1}{r_2} e^{i(\theta_1 - \theta_2)}, \quad z_2 \neq 0$$

~~In particular, $z^{-1} = r^{-1} e^{-i\theta}$, $z \neq 0$~~

Lemma: For every $n \in \mathbb{Z}$, $z^n = r^n e^{in\theta}$. Here $z = re^{i\theta}$, $z \neq 0$.
($z \neq 0$) For $k \in \mathbb{N}$, $z^{-k} = (z^{-1})^k$.

Proof: for $n \geq 1$, we prove using mathematical induction.

sp. $z^m = r^m e^{im\theta}$ for $m \geq 1$, then

$$\begin{aligned} z^{m+1} &= z \cdot z^m = re^{i\theta} \cdot r^m e^{im\theta} \\ &= r^{m+1} e^{i(m+1)\theta} \end{aligned}$$

Thus, $z^n = r^n e^{in\theta}$ $\forall n \geq 1$ & by verification for $n=0$.

Now, consider $\frac{1}{z}$ & z^m for $m < 0$. Then,

$$z^m = (z^{-1})^{|m|} = (r^{-1} e^{-i\theta})^{|m|} = r^m e^{im\theta}$$

True for every $m < 0$.

Exponential form can help in finding powers of complex nos.

$$\text{Eg 1. } (\sqrt{3}+i)^7 = 2^7 \left(\frac{\sqrt{3}}{2} + i \times \frac{1}{2}\right)^7 = 2^7 (e^{i\pi/6})^7 = 2^7 e^{7\pi i/6} = -2^7 e^{5\pi i/6}$$

Arguments of products & quotients

If $z_j = r_j e^{i\theta_j}$ for $j=1,2$, then

$$\arg(z_1 z_2) = \arg z_1 + \arg z_2. \text{ Here, } \arg z_j = \{\theta_j + 2m_j\pi \mid m_j \in \mathbb{Z}\}.$$

Proof:- Note: $\theta_1 \in \arg z_1$ & $\theta_2 \in \arg z_2$. We prove $\text{LHS} \subseteq \text{RHS}$ & $\text{RHS} \subseteq \text{LHS}$

$$\text{Now, let } \theta \in \arg(z_1 z_2) \Rightarrow z_1 z_2 = |z_1 z_2| e^{i\theta} = r_1 r_2 e^{i\theta}$$

$$\text{But } z_1 z_2 = r_1 e^{i\theta_1} r_2 e^{i\theta_2} \Rightarrow \theta_1 + \theta_2 = \theta + 2n\pi \text{ for some } n \in \mathbb{Z}$$

$$\Rightarrow \theta \in \arg z_1 + \arg z_2 \quad \Rightarrow \theta = (\theta_1 - 2n\pi) + (\theta_2) \in \arg z_1 + \arg z_2$$

$$\text{Similarly, let } \alpha_j \in \arg z_j \Rightarrow \alpha_1 = \theta_1 + 2n_1\pi \text{ & } \alpha_2 = \theta_2 + 2n_2\pi$$

$$\Rightarrow \alpha_1 + \alpha_2 = \theta_1 + \theta_2 + 2\pi(n_1 + n_2)$$

$$\in \arg(z_1 z_2)$$

$$\approx r_1 r_2 e^{i(\theta_1 + \theta_2)} = z_1 z_2$$

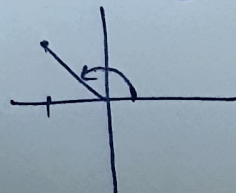
Qn 1) Is $\text{Arg } z_1 + \text{Arg } z_2 = \text{Arg}(z_1 z_2)$?

(Here, $-\pi < \text{Arg } z \leq \pi$). If $z_1 = e^{i\pi} = -1$ & $z_2 = e^{i\pi/2} = i$, then

$$\text{Arg } z_1 = \pi \text{ & } \text{Arg } z_2 = \frac{\pi}{2} \Rightarrow \text{Arg } z_1 + \text{Arg } z_2 = \frac{3\pi}{2}$$

$$\text{but } \text{Arg}(z_1 z_2) = \text{Arg}(-i)$$

$$= -\frac{\pi}{2}$$



Ex 1) $z \neq 0$, $\arg z^{-1} = -\arg z$

From prev. lemma, if $z_2 \neq 0$

$$\arg\left(\frac{z_1}{z_2}\right) = \arg z_1 - \arg z_2$$

True if $\text{Re}(z_1) > 0$ & $\text{Re}(z_2) > 0$.

Eg 2) Let $z = \frac{-2}{1+\sqrt{3}i}$, find $\text{Arg } z$.

(3)

~~Let~~ Let $z_1 = -2$ & $z_2 = 1+\sqrt{3}i$

$$\arg\left(\frac{z_1}{z_2}\right) = \arg z_1 - \arg z_2$$

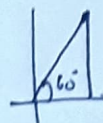
$$\arg z_1 = \{\pi + 2n\pi \mid n \in \mathbb{Z}\}$$

$$-\arg z_2 = \left\{-\frac{\pi}{3} + 2m\pi \mid m \in \mathbb{Z}\right\}$$

$$\Rightarrow \arg z = \left\{\frac{2\pi}{3} + 2l\pi \mid l \in \mathbb{Z}\right\}$$

Clearly $\text{Arg}\left(\frac{z_1}{z_2}\right)$ is the unique elt. of $\arg\left(\frac{z_1}{z_2}\right)$ s.t. $-\pi < \text{Arg}\left(\frac{z_1}{z_2}\right) \leq \pi$

$$\Rightarrow \text{Arg } z = \frac{2\pi}{3}$$



Lee 15

Roots of complex nos.

$\sqrt{\cdot} : \mathbb{R}^+ \rightarrow \mathbb{R}^+$ cannot be taken as $-\sqrt{\cdot}$.
Let $r_0 > 0$. Then, n^{th} root of r_0 is that unique $+ \text{no. } r \text{ s.t. } r^n = r_0$. For eg. square root of 3 is $\sqrt{3}$.

Let $z_0 = r_0 e^{i\theta_0}$ where $z_0 \neq 0$ & we aim to find the n^{th} roots of z_0 . ie, ~~a non-zero~~ ^{all} z s.t. $z^n = z_0$

($\because |z^n| = |z_0|$, it follows that $z \neq 0$ & hence, $z = re^{i\theta}$)

$$\text{ie, } r^n e^{in\theta} = r_0 e^{i\theta_0}$$

$$\therefore r^n = r_0 \text{ \& } n\theta = \theta_0 + 2k\pi, \text{ for some } k \in \mathbb{Z}.$$

$$\text{ie, } r = \sqrt[n]{r_0} \text{ \& } \theta = \frac{\theta_0}{n} + \frac{2k\pi}{n}, \text{ } k \in \mathbb{Z}. \quad \text{--- (1)}$$

(the unique n^{th} root of r_0 .)

$\therefore$ Any z of the form $re^{i\theta}$ where r & θ are as in

(1) ~~is~~ merely satisfies $z^n = z_0$ & hence, all the complex nos. of

$$\text{the form } z = \sqrt[n]{r_0} e^{i\left(\frac{\theta_0}{n} + \frac{2k\pi}{n}\right)} \text{ where } k \in \mathbb{Z}$$

are the n^{th} roots of unity.

But many of them repeat. For eg,

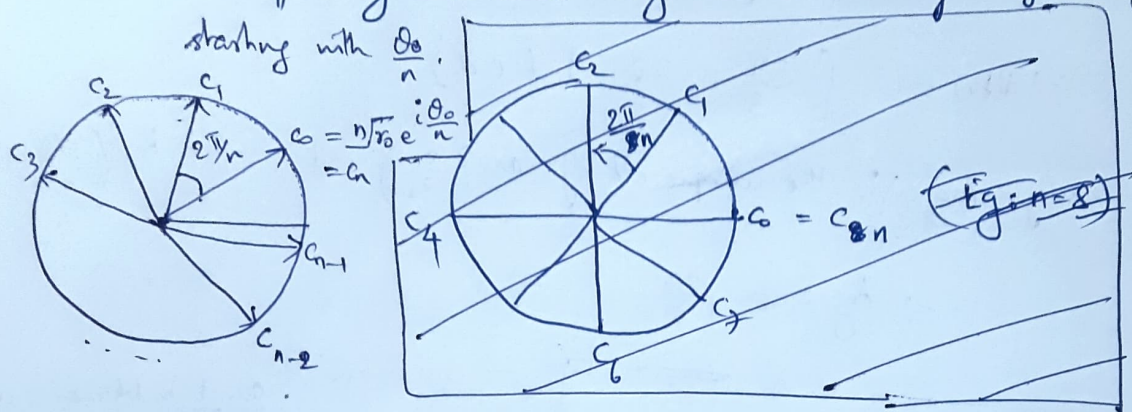
$$c_k = \sqrt[n]{r_0} e^{i\left(\frac{\theta_0}{n} + \frac{2k\pi}{n}\right)} = c_{k+n} = \sqrt[n]{r_0} e^{i\left(\frac{\theta_0}{n} + \frac{2(k+n)\pi}{n}\right)}$$

Thus, the distinct n^{th} roots of z_0 are given by

$$c_k = \sqrt[n]{r_0} e^{i\left(\frac{\theta_0}{n} + \frac{2k\pi}{n}\right)} \quad \text{where } 0 \leq k \leq n-1.$$

There are n many distinct n^{th} roots of z_0 . We can plot them on

the circle $|z| = \sqrt[n]{r_0}$ at about the origin & they are equally distributed angular in terms of angles by an angle $\frac{2\pi}{n}$ starting with $\frac{\theta_0}{n}$.



$\forall 0 \leq k \leq n-1$, length of $c_k = \sqrt[n]{r_0}$

Where $n \geq 3$, the roots lie at the vertices of a regular n gon inscribed in that circle. We define the following symbol for the set of n^{th} roots of z_0 :

$$\text{Let } z_0^{1/n} = \{c_k \mid 0 \leq k \leq n-1\}$$

If $z_0 \in \mathbb{R}^+$, i.e. $z_0 = r_0$, then

$(r_0)^{1/n}$ denotes the whole set of n roots of r_0

& $\sqrt[n]{r_0}$ denotes the unique n^{th} root of r_0 .

For e.g. $(2)^{1/3}$ & $\sqrt[3]{2}$ are different for us from now onwards
like $\arg i$ & $\frac{\pi}{2}$ are $\sqrt[3]{2} \in (2)^{1/3}$.

$$\sqrt[3]{2} \in (2)^{1/3} \quad \text{like} \quad \frac{\pi}{2} \in \arg i$$

In the expression

$$c_k = \sqrt[n]{r_0} e^{i\left(\frac{\theta_0}{n} + \frac{2k\pi}{n}\right)}$$

if $-\pi < \theta_0 \leq \pi$, then c_0 is called the principal root of z_0 .

Thus, if $z_0 \in \mathbb{R}^+$, ~~say~~ $z_0 = r_0$, then

$$c_0 = \sqrt[n]{r_0}$$

ie, the n th root of r_0 is the principal root of r_0 .

Consider

$$c_k = \sqrt[n]{r_0} e^{i \frac{\theta_0}{n}} e^{i \frac{2k\pi}{n}}$$

Let $\omega_n = e^{i \frac{2\pi}{n}}$. Then

$$c_k = \sqrt[n]{r_0} e^{i \frac{\theta_0}{n}} (\omega_n)^k$$

Further, $c_0 = \sqrt[n]{r_0} e^{i \frac{\theta_0}{n}}$

$$\Rightarrow \boxed{c_k = c_0 (\omega_n)^k} \quad \text{for each } 0 \leq k \leq n-1.$$

Eg 5) Find all cube roots of $(-8i)$.

$z_0 = -8i = r_0 e^{i\theta_0}$. Here, $r_0 = 8$ & $e^{i\theta_0} = -i$
 we choose for simplicity, ~~let~~ $\theta_0 = -\frac{\pi}{2}$.

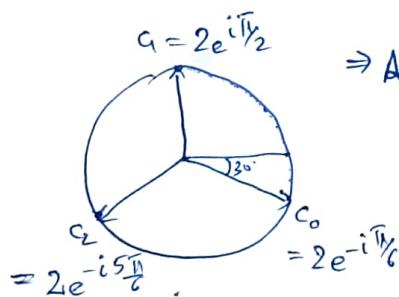
$$\Rightarrow z_0 = 8 \times e^{-i\pi/2}$$

Note $c_k = c_0 (\omega_3)^k$ where $c_0 = (r_0)^{1/3} e^{i \frac{\theta_0}{3}}$
 $= 2 e^{-i\pi/6}$

$$\Rightarrow \text{Ans} = \left\{ 2 e^{-i\pi/6}, 2 e^{-i\pi/6} e^{i2\pi/3}, 2 e^{-i\pi/6} e^{i4\pi/3} \right\}$$

ie, $c_k = 2 e^{-i\pi/6} (e^{i2\pi/3})^k$; $k=0,1,2$

$$\text{Ans} = \{c_0, c_0 \omega_3, c_0 (\omega_3)^2\}$$



Exercise: a) n th root of unity, ie 1 are $\{1, \omega_n, (\omega_n)^2, \dots, (\omega_n)^{n-1}\}$
 $= \{e^{2\pi i/n \times k} \mid 0 \leq k \leq n-1\}$.

Q2) ~~Show that if c is~~ Establish the identity

$$1 + z + z^2 + \dots + z^n = \frac{1 - z^{n+1}}{1 - z}, \quad z \neq 1. \quad (6)$$

& use it to derive Lagrange trigonometric identity

$$1 + \cos \theta + \cos 2\theta + \dots + \cos n\theta = \frac{1}{2} + \frac{\sin \left[\frac{(2n+1)\theta}{2} \right]}{2 \sin \left(\frac{\theta}{2} \right)}, \quad 0 < \theta < 2\pi$$

Q3) Show that if c is any n^{th} root of unity other than unity itself, ⁽¹⁾
then $1 + c + c^2 + \dots + c^{n-1} = 0$.

Q4) Prove that the usual formula solves the quadratic eqn

$$az^2 + bz + c = 0 \quad a \neq 0$$

when the coefficients a, b & c are complex nos. Specifically, by completing the square on the left-hand side, derive the quadratic formula

$$z = \frac{-b \pm (b^2 - 4ac)^{1/2}}{2a}$$

where both square roots are to be considered when $b^2 - 4ac \neq 0$.

Q5) Find the four zeros of the poln. $z^4 + 4$ & ~~use~~ factorise it into

Important Formulas

for $A, B \in \mathbb{R}, \theta \in \mathbb{R}$,

quadratic factor
with real coefficients.

$$1) \cos(A+B) = \cos A \cos B - \sin A \sin B$$

$$2) \cos(A-B) = \cos A \cos B + \sin A \sin B$$

$$3) \cos A + \cos B = 2 \cos \left(\frac{A+B}{2} \right) \cos \left(\frac{A-B}{2} \right)$$

$$4) \cos A - \cos B = -2 \sin \left(\frac{A+B}{2} \right) \sin \left(\frac{A-B}{2} \right)$$

$$5) \cos^2 \left(\frac{\theta}{2} \right) = \frac{1 + \cos \theta}{2} \quad \& \quad \sin^2 \left(\frac{\theta}{2} \right) = \frac{1 - \cos \theta}{2}$$

Use 5 to prove 3 & 4.

$$6) \sin A - \sin B = 2 \cos \left(\frac{A+B}{2} \right) \sin \left(\frac{A-B}{2} \right)$$

$$7) \sin A + \sin B = 2 \sin \left(\frac{A+B}{2} \right) \cos \left(\frac{A-B}{2} \right).$$

(7)

Q2: Soln

$$a) \text{ let } S(z) = 1 + z + z^2 + \dots + z^n$$

$$\Rightarrow z S(z) = z + z^2 + \dots + z^n + z^{n+1}$$

$$\Rightarrow (1-z) S(z) = 1 - z^{n+1} \Rightarrow S(z) = \frac{1 - z^{n+1}}{1-z}, \quad z \neq 1$$

$$b) z = e^{i\theta} \rightarrow$$

$$1 + e^{i\theta} + \dots + e^{in\theta} = \frac{1 - e^{i(n+1)\theta}}{1 - e^{i\theta}} = \frac{[1 - e^{i(n+1)\theta}][1 - e^{-i\theta}]}{|1 - \cos\theta - i\sin\theta|^2}$$

$$= \frac{1 - e^{i(n+1)\theta} - e^{-i\theta} + e^{in\theta}}{(1 - \cos\theta)^2 + (\sin\theta)^2}$$

$$\Rightarrow \frac{1 + \cos\theta + \cos 2\theta + \dots + \cos n\theta}{1 - \cos\theta} \Rightarrow$$

$$= \frac{1 - \cos((n+1)\theta) - \cos\theta + \cos n\theta}{(1 - \cos\theta)^2 + (\sin\theta)^2}$$

$$1 - \cos\theta = 2 \sin^2 \frac{\theta}{2} \Rightarrow \text{denom} = 4 \sin^4 \frac{\theta}{2}$$

$$(1 - \cos\theta)^2 + \sin^2\theta = 2 - 2\cos\theta = 4 \sin^2 \frac{\theta}{2}$$

$$\therefore \sum_{k=0}^n \cos(k\theta) = \frac{2 \sin^2 \frac{\theta}{2} + \cos n\theta - \cos((n+1)\theta)}{4 \sin^2 \frac{\theta}{2}}$$

$$= \frac{1}{2} + \frac{-2 \sin\left(\frac{n\theta + (n+1)\theta}{2}\right) \sin\left(\frac{n\theta - (n+1)\theta}{2}\right)}{4 \sin^2 \frac{\theta}{2}}$$

$$= \frac{1}{2} + \frac{\sin\left[\frac{(2n+1)\theta}{2}\right]}{2 \sin \frac{\theta}{2}}; \quad 0 < \theta < 2\pi$$

Q3) If $c \in (1)^n \setminus \{1\}$, then

(8)

$$1 + c + \dots + c^{n-1} = \frac{1 - c^n}{1 - c} = 0$$

Q4) $az^2 + bz + c = a \left(z^2 + \frac{b}{a}z + \frac{c}{a} \right) = a \left(\left(z + \frac{b}{2a} \right)^2 - \frac{b^2}{4a^2} + \frac{c}{a} \right)$

$$= a \left[\cancel{z + \frac{b}{2a}} \left(z + \frac{b}{2a} \right)^2 - \left(\frac{b^2 - 4ac}{4a^2} \right) \right] = 0$$

$$a \neq 0 \Rightarrow \left(z + \frac{b}{2a} \right)^2 = \left(\frac{b^2 - 4ac}{4a^2} \right)^{1/2}$$

$$\therefore z \in -\frac{b}{2a} + \left(\frac{b^2 - 4ac}{4a^2} \right)^{1/2}$$

Lee 16:

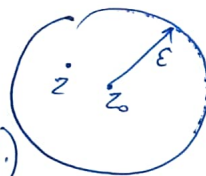
Regions in the Complex plane

Let $\epsilon > 0$ & $z_0 \in \mathbb{C}$. ~~Then~~ and $S \subseteq \mathbb{C}$. Then,

Defn 1) ϵ -neighbourhood (of z_0)

$$= \{ z \in \mathbb{C} \mid |z - z_0| < \epsilon \}$$

(written also as $B(z_0, \epsilon)$ or $B_\epsilon(z_0)$.)



2) deleted neighbourhood / punctured disk

$$= \{ z \in \mathbb{C} \mid 0 < |z - z_0| < \epsilon \}$$

3) Interior pt of S

A pt z in $\mathbb{C}$ is said to be an interior point of S whenever there is some neighbourhood of z that contains only points of S , i.e., $\exists \epsilon > 0$ s.t. $B(z_0, \epsilon) \subseteq S$.

4) Exterior pt of S

... whenever there is some nbhd of z_0 that contains no points of S i.e., $\exists \epsilon > 0$ s.t. $B(z_0, \epsilon) \subseteq S^c$.

- 5) If z_0 is neither an interior point nor an exterior point, (9)
then it is called a boundary point of S .

Hence, equivalently, a point all of whose neighbourhoods contain at least one point in S & at least one point in S^c is called a boundary point.

6) Boundary of S , ∂S

The collection of all boundary points is called the boundary of S .

- 7) ~~A set is called open if it contains~~ Interior of S , $\text{int}(S)$

The collection of all interior points of S .

- 8) A set S is open if every pt in S is an interior point of S .

- 9) A set S is closed if its complement S^c (in $\mathbb{C}$) is open.

- 10) A point $z \in \mathbb{C}$ is said to be ^(accumulation point) a limit point of S if $\exists z_n \in S$ such

(Sequential defn) ~~that~~ a sequence of points $\{z_n\}_{n=1}^{\infty}$ ($z_n \in S$) such that $z_n \neq z$ &
 $\lim_{n \rightarrow \infty} z_n = z$. (ii) equivalently, if each deleted neighbourhood of z contains
one pt of S . (Ex 1)

Exercise 2) S is closed $\Leftrightarrow S$ contains all its limit points.

- 11) Closure of S , $\bar{S}$ is the union of S & its limit points.

Exercise 3) $\partial S = \bar{S} \setminus \text{int}(S)$

- 12) A set S is bounded if $\exists R > 0$ such that $|z| < R \forall z \in S$.
if no such $R > 0$ exists, then S is unbounded.

- 13) An open set $S \subseteq \mathbb{C}$ is ~~connected~~ ^{disconnected} if there exists two disjoint non-empty open sets S_1 & S_2 such that
 $S = S_1 \cup S_2$

An open set is connected if it is not disconnected.

- 14) A connected open set in $\mathbb{C}$ is called a region or sometimes a domain.

Theorem

(10)

If z_0 is a limit point of S , then every neighbourhood of z_0 contains only many pts of S .

Corollary

A finite set has no limit points.

~~Ex 2)~~ Ex 2) S is closed $\Leftrightarrow S$ contains all its limit points.

" $\Rightarrow$ " Let z_0 be a limit pt of S . Then, $\exists \epsilon > 0$ s.t.

$$\{0 < |z - z_0| < \epsilon\} \cap S \neq \emptyset$$

If $\forall z \notin S$, then $z \in S^c \Rightarrow z_0$ is an int pt of S^c as S^c is open.

$$\Rightarrow \exists \epsilon > 0 \text{ s.t. } B(z_0, \epsilon) \subseteq S^c$$

$$\Rightarrow \therefore B(z_0, \epsilon) \cap S = \emptyset$$

$\rightarrow$ as z_0 is a limit pt of S .

$\Leftarrow$ Let S contains all its limit pts. Then, we prove S^c is open.

On the contrary, if S^c was not open, $\exists z \in S^c$ which is not an interior pt of $S^c \Rightarrow \forall \epsilon > 0, B(z_0, \epsilon) \cap S \neq \emptyset$

$\therefore z \notin S, z_0 \therefore z$ has the property that every deleted nbd $B(z_0, \epsilon) \cap S \neq \emptyset$ as $z \notin S$. Hence, z is a limit pt of S

But then $z \in S$ as S contains all its limit pts.

$$\therefore z \in S \text{ \& } z \in S^c \text{ } \rightarrow \text{ } \rightarrow$$

Functions of a complex variable $S \subseteq \mathbb{C}$

~~Let $w = f(z)$~~ A function f defined on S is a rule that assigns to each $z \in S$ a complex no. w , i.e., $w = f(z)$

$\exists$ ^{that} $w = u + iv$ ~~is~~ is the value of f at $z = x + iy$, k.s. that $u + iv = f(x + iy)$.

Here, each real no. u & v depend on x & y .

$$\text{i.e. } f(z) = u(x, y) + iv(x, y).$$

If $z = z_0$ is written as $re^{i\theta}$, then, $u + iv = f(re^{i\theta})$

$$\& u = u(r, \theta) \& v = v(r, \theta). \& f(z) = u(r, \theta) + iv(r, \theta).$$

Functions on the complex plane

(11)

Let f be a function defined on a set at all points z in some deleted neighbourhood of z_0 .

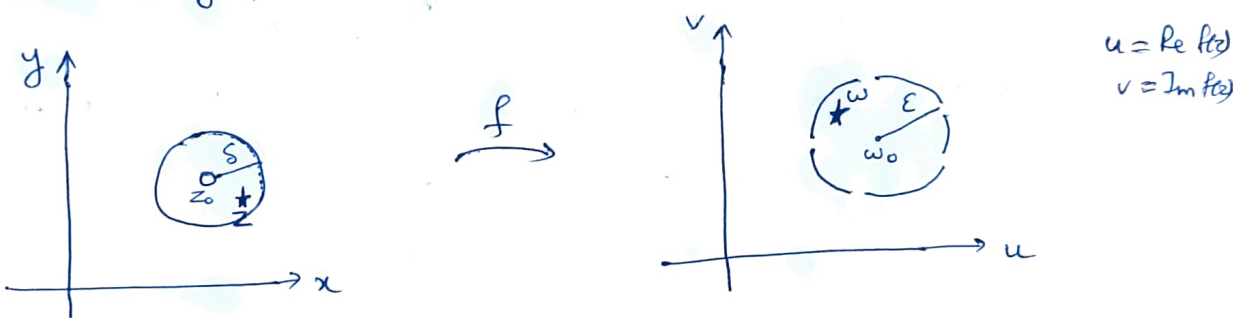
$\lim_{z \rightarrow z_0} f(z) = w_0$ means that the point $w = f(z)$ can be made

arbitrarily close to w_0 if we choose the point z close enough to z_0 but distinct from it.

Equivalently, for each $\epsilon > 0$, $\exists \delta > 0$ s.t.

(2) $|f(z) - w_0| < \epsilon$ whenever $0 < |z - z_0| < \delta$.

Geometrically, this says that for each ϵ nbhd of w_0 , there is a deleted δ neighbourhood of z_0 such that every point z in it has an image w lying in the ϵ nbhd of w_0 .



PS: it is not necessary that every pt of $B(w_0, \epsilon)$ has a pre-image in $B^{\text{del}}(z_0, \delta)$.

e) For given $\epsilon > 0$, if some $\delta > 0$ satisfies $\star$, so does $\frac{\delta}{2}$, etc....

Lemma: When a $\lim_{z \rightarrow z_0} f(z) = w_0$ exists, w_0 is said to be the limit of $f(z)$ as z approaches z_0 .

Lemma: When a limit of a function $f(z)$ exists at a point z_0 , it is unique.

Proof: Let $\lim_{z \rightarrow z_0} f(z) = w_1$ & $\lim_{z \rightarrow z_0} f(z) = w_2$. Then,

$$|w_1 - w_2| = |w_1 - f(z) + f(z) - w_2| \quad \forall z \text{ in domain of defn of } f$$

(12)

Note that by defn, for any given $\epsilon > 0$, $\exists \delta_1 > 0$ st

$$|f(z) - w_1| < \epsilon \quad \text{whenever} \quad 0 < |z - z_0| < \delta_1$$

& $\exists \delta_2 > 0$ st

$$|f(z) - w_2| < \epsilon \quad \text{whenever} \quad 0 < |z - z_0| < \delta_2$$

$$\therefore |w_1 - w_2| \leq 2\epsilon \quad \text{holds for any } \epsilon > 0$$

$$\text{Hence, } w_1 = w_2$$

Defn: A limit point of a set is either in $\mathbb{R}^n$ or $\mathbb{C}$.
~~Defn (1) or (2) requires that~~

If z_0 is an interior point of S , then ~~also~~ such a deleted nbhd of z_0 always exists inside S .

If z_0 is a boundary pt of S , then we modify (2) or (1) as follows

Let z_0 be a boundary pt of S ~~is said~~. ~~f is said to have limit as z tends to~~

~~The limit of $f(z)$ as z approaches z_0 is w_0~~

$$\text{Lt}_{z \rightarrow z_0} f(z) = w_0 \text{ means } \forall \epsilon > 0, \exists \delta > 0 \text{ st}$$

$$|f(z) - w_0| < \epsilon \quad \text{whenever } z \text{ lies in } S \text{ \& satisfies}$$

Let $f: S \rightarrow \mathbb{C}$ & let z_0 be a limit pt of S . We say
 Lt $f(z) = L$ provided that for all $\epsilon > 0$, $\exists \delta > 0$ such that
 $z \rightarrow z_0$ whenever $0 < |z - z_0| < \delta$ (& $z \in S$), $|f(z) - L| < \epsilon$

$$\text{Eg 1) Let } f(z) = i\frac{z}{2}, \quad |z| < 1. \quad \text{Then, } \lim_{z \rightarrow 1} f(z) = \frac{i}{2}.$$

Proof: Let $\epsilon > 0$ be any arbitrary re no. We need to show that $\exists \delta > 0$ st

$$0 < |z - 1| < \delta \Rightarrow |f(z) - \frac{i}{2}| < \epsilon.$$

$$\text{Spz } |f(z) - \frac{i}{2}| < \epsilon, \text{ ie, } \left| i\frac{z}{2} - \frac{i}{2} \right| = \frac{|z-1|}{2} = \frac{|x-1|}{2} = \frac{|x-1|}{2} \left| \frac{z-1}{z-1} \right| \left| \frac{z-1}{z-1} \right|$$

Eg 2) Let $f(z) = \frac{z}{z}$. Show that Lt $f(z)$ doesn't exist.
 For any ϵ

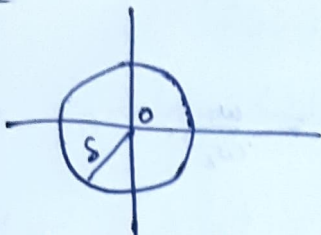
Clearly if $z \in \mathbb{R}$, Lt $f(z) = 1$ & if $z = ix, x \in \mathbb{R}$, Lt $f(z) = \frac{ix}{-ix} = -1$

Thus, we need to prove by contradiction.

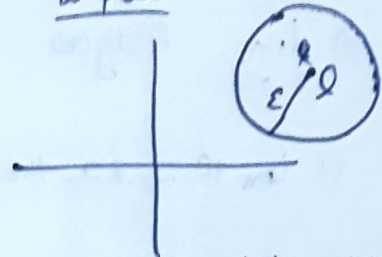
Suppose $\lim_{z \rightarrow 0} f(z) = l$. Then, for any given $\epsilon > 0$, $\exists \delta > 0$ s.t.

$$0 < |z| < \delta \Rightarrow |f(z) - l| < \epsilon \quad (*)$$

z plane $\Rightarrow z \in B_\delta^{\text{del}}(0) \Rightarrow f(z) \in B_\epsilon(l, \epsilon)$
 ω plane



f



To contradict,

We need to show that there is an $\epsilon > 0$ for which even no δ satisfies $(*)$

Clearly any deleted neighbourhood of 0 contains ~~the~~ pts on real axis & ~~other~~ points i on the imaginary pts.

Let $\epsilon < 1$. Let $z_1 = \frac{\delta}{2}$ & $z_2 = \frac{i\delta}{2}$

$$|f(z_1) - f(z_2)| = |f(z_1) - l + l - f(z_2)| \leq |f(z_1) - l| + |f(z_2) - l| < 2\epsilon$$

However, as $f(z_1) = 1$ & $f(z_2) = -1$, $|f(z_1) - f(z_2)| = 2$.

Hence, hence, ~~one assumption was~~ no such δ can exist, given any $\epsilon > 0$, ~~no such~~ no such δ can exist.

$\therefore \lim_{z \rightarrow 0} f(z)$ doesn't exist.

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Theorem 1) Suppose that $f(z) = u(x, y) + iv(x, y)$ where $z = x + iy$,

$z_0 = x_0 + iy_0$ & $u_0 = u(x_0, y_0)$. Then,

$$\lim_{z \rightarrow z_0} f(z) = w_0 \text{ iff}$$

$$\lim_{(x, y) \rightarrow (x_0, y_0)} u(x, y) = u_0 \quad \text{and} \quad \lim_{(x, y) \rightarrow (x_0, y_0)} v(x, y) = v_0.$$

Recall: Meaning: $\lim_{(x, y) \rightarrow (x_0, y_0)} \|u(x, y) - u_0\| = 0$.

$$\|(x, y) - (x_0, y_0)\| \rightarrow 0$$

Theorem 2 (Algebra on limits) Suppose

$$\lim_{z \rightarrow z_0} f(z) = w_1 \quad \& \quad \lim_{z \rightarrow z_0} g(z) = w_2 \quad \text{Then,}$$

$$1) \lim_{z \rightarrow z_0} (f(z) + g(z)) = w_1 + w_2$$

$$2) \lim_{z \rightarrow z_0} f(z)g(z) = w_1 w_2$$

$$3) \text{ If } w_2 \neq 0, \text{ then } \lim_{z \rightarrow z_0} \frac{f(z)}{g(z)} = \frac{w_1}{w_2}$$

Continuity

A function $f: S \rightarrow \mathbb{C}$ is continuous at $z_0 \in S$ if

$$\lim_{z \rightarrow z_0} f(z) = f(z_0)$$

Note that ~~this~~ it is implicit in this defn that

$$1) \lim_{z \rightarrow z_0} f(z) \text{ exists}$$

$$2) f(z_0) \text{ exists}$$

In other words, for each $\epsilon > 0$, $\exists \delta > 0$ s.t.

$$|f(z) - f(z_0)| < \epsilon \quad \text{whenever} \quad |z - z_0| < \delta$$

Further $\lim_{z \rightarrow z_0} f(z) = f(z_0) = f(\lim_{z \rightarrow z_0} z)$ i.e. limit & f can be interchanged if f is continuous at z_0 .

f is continuous in a region R if it is continuous at each point in R .

FACT 1) If f & g are continuous at $z_0 \in S$, $(f, g: S \rightarrow \mathbb{C})$, then

$$f+g, fg, \frac{f}{g} \text{ (provided } g(z_0) \neq 0 \text{)} \text{ are}$$

also continuous at z_0 .

2) A composition of two ~~funct~~ continuous functions is also continuous.

3) If a function $f(z)$ is continuous and non-zero at a point z_0 , then $f(z) \neq 0$ throughout some neighbourhood of z_0 .

4) The function $f(z) = u(x,y) + iv(x,y)$ is continuous at a point $z = x_0 + iy_0$ iff $u(x,y)$ & $v(x,y)$ are continuous at (x_0, y_0) . (15)

Theorem 3 If a function $f(z)$ is continuous throughout a region R that is both closed & bounded, then $\exists M \geq 0$ such that

$$|f(z)| \leq M \quad \forall z \in R$$

where equality holds for atleast one such z .

Exercise 1: ~~What is the~~ ^{Discuss the limiting behaviour} ~~show that the limit of the function~~

$$f(z) = \left(\frac{z}{\bar{z}}\right)^2 \text{ as } z \rightarrow 0 \text{ ~~doesn't exist~~. (Hint: Line } y=x)$$

Derivatives

Let $z_0 \in \mathbb{C}$.

Let f be a function whose domain of defn contains a neighbourhood $|z - z_0| < \epsilon$ of the pt z_0 . The derivative of f at z_0 is the limit

$$\lim_{z \rightarrow z_0} \frac{f(z) - f(z_0)}{z - z_0}$$

if it exists. If the above limit exists, we denote it by $f'(z_0)$ & say that " f is differentiable at z_0 ".

If $\Delta z = z - z_0$, then equivalently, we may also write

$$f'(z_0) \text{ as } \lim_{\Delta z \rightarrow 0} \frac{f(z_0 + \Delta z) - f(z_0)}{\Delta z}$$

Since f is defined on some $B(z_0, \epsilon)$, ($f(z_0 + \Delta z)$ is defined on pts $z_0 + \Delta z \in B(z_0, \epsilon)$ &) hence, ~~limit can be~~ ^{is well defined} the expression $\lim_{\Delta z \rightarrow 0} \frac{f(z_0 + \Delta z) - f(z_0)}{\Delta z}$ makes sense.

Eg. 1) $f(z) = \bar{z}$ is NOT differentiable anywhere on $\mathbb{C}$.

$$\lim_{\Delta z \rightarrow 0} \frac{f(z_0 + \Delta z) - f(z_0)}{\Delta z} = \lim_{\Delta z \rightarrow 0} \frac{\bar{z_0 + \Delta z} - \bar{z_0}}{\Delta z} \text{ doesn't exist as}$$

seen in Eg 2) Limits class.

2) $f(z) = |z|^2 = z\bar{z}$

$$\lim_{h \rightarrow 0} \frac{f(z+h) - f(z)}{h} = \lim_{h \rightarrow 0} \frac{z\bar{h} + \bar{z}h + h\bar{h}}{h}$$

Along real axis, ($h \in \mathbb{R}$), this limit exists & equals $z + \bar{z}$

Along imaginary axis, ($h = ik, k \in \mathbb{R}$), this limit exists &

equals $\lim_{k \rightarrow 0} \left(z \frac{-ik}{ik} + \bar{z} \frac{ik}{ik} + -ik \right) = \bar{z} - z$

Thus, ~~for~~ f has no ~~limit~~ derivative anywhere on $\mathbb{C}$ ^{except} ~~unless~~

at $z=0$. (Note: $z + \bar{z} = \bar{z} - z \iff 2z = 0 \iff z = 0$)

This is an example of a function $f(z) = |z|^2$ which is ^{complex} differentiable at a single point & nowhere else in any nbhd of that point.

Usual identities involving derivatives of real valued functions work here also:-

1) $\frac{d}{dz}(c) = 0, c \in \mathbb{C}$

2) $\frac{d}{dz}(z^n) = nz^{n-1}, n \in \mathbb{Z}$

3) $\frac{d}{dz}(z^n) = nz^{n-1}, n \in \mathbb{Z}, z \neq 0$

4) Sum rule

5) Product rule

6) Quotient rule

7) Chain rule: If f has deriv. at z_0 & g has deriv. at $f(z_0)$. Then

the function $F(z) = g[f(z)]$ has derivative at z_0 &

$$F'(z_0) = g'[f(z_0)] f'(z_0).$$

Cauchy-Riemann Equations

AIM: To obtain alternate ways to talk about complex differentiability.

$$\text{Let } f(z) = u(x, y) + iv(x, y) \quad \& \quad z = x_0 + iy_0.$$

$$\text{Let } \Delta z = \Delta x + i\Delta y = h + ik$$

$$\begin{aligned} \Delta w &= f(z_0 + \Delta z) - f(z_0) \\ &= u(x_0 + \Delta x, y_0 + \Delta y) - u(x_0, y_0) + i[v(x_0 + \Delta x, y_0 + \Delta y) - v(x_0, y_0)] \end{aligned}$$

Recall

$$f'(z_0) = \lim_{\Delta z \rightarrow 0} \frac{f(z_0 + \Delta z) - f(z_0)}{\Delta z} = \lim_{\Delta z \rightarrow 0} \frac{\Delta w}{\Delta z} = \lim_{\Delta z \rightarrow 0} \frac{\Delta w}{\Delta z}$$

$$= \lim_{(\Delta x, \Delta y) \rightarrow (0, 0)} \frac{\Delta w}{\Delta z} \quad \text{Recall that if } \lim_{z \rightarrow z_0} f(z) \text{ exists is}$$

$$\text{equivalent to saying } \lim_{(x, y) \rightarrow (x_0, y_0)} u(x, y) + i \lim_{(x, y) \rightarrow (x_0, y_0)} v(x, y)$$

$$\therefore f'(z_0) = \lim_{(\Delta x, \Delta y) \rightarrow (0, 0)} \operatorname{Re} \left[\frac{\Delta w}{\Delta z} \right] + i \lim_{(\Delta x, \Delta y) \rightarrow (0, 0)} \operatorname{Im} \left[\frac{\Delta w}{\Delta z} \right].$$

Theorem 4) Suppose that $f(z) = u(x, y) + iv(x, y)$ & that $f'(z)$ exists at a point $z = x_0 + iy_0$. Then, the first order partial derivatives

1) the first order partial derivatives of u & v exist at (x_0, y_0) .

2) first order partial derivatives must satisfy the CR eqns

$$u_x = v_y \quad \& \quad u_y = -v_x.$$

3) further,

$$f'(z_0) = u_x + iv_x \Big|_{(x_0, y_0)}$$

Ex 1) $f(z) = z^2 \Rightarrow u(x, y) = x^2 - y^2 \quad \& \quad v(x, y) = 2xy$
 $= (x+iy)^2$
 $= x^2 - y^2 + i2xy$
 $\& \quad f'(z) = 2z = u_x + iv_x$

$$\begin{aligned} u_x &= 2x & v_y &= 2x \\ u_y &= -2y & v_x &= 2y \end{aligned}$$

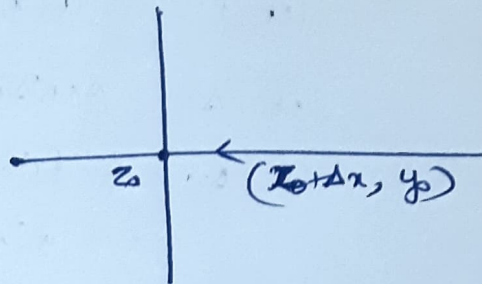
Proof: Since $f'(z_0)$ exists, the limit

$$\lim_{\Delta z \rightarrow 0} \frac{f(z) - f(z_0)}{\Delta z} \quad \lim_{\Delta z \rightarrow 0} \frac{f(z_0 + \Delta z) - f(z_0)}{\Delta z} \text{ exists \& hence is}$$

same along all directions.

Consider along X axis :-

$$\lim_{\Delta x \rightarrow 0} \frac{f(z_0 + \Delta x + iy_0) - f(z_0 + iy_0)}{\Delta x}$$



$$= \lim_{\Delta x \rightarrow 0} \frac{u(x_0 + \Delta x, y_0) - u(x_0, y_0)}{\Delta x} + i \frac{v(x_0 + \Delta x, y_0) - v(x_0, y_0)}{\Delta x}$$

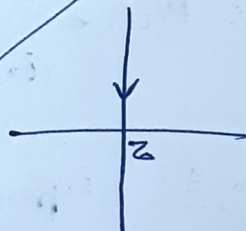
$$= \frac{\partial u}{\partial x}(x_0, y_0) + i \frac{\partial v}{\partial x}(x_0, y_0)$$

both these exist from Thm 1

ie, $f'(z_0) = u_x(x_0, y_0) + i v_x(x_0, y_0)$ (Part 2 is proved)

Consider along Y axis :-

$$\lim_{\Delta y \rightarrow 0} \frac{f(x_0 + i(y_0 + \Delta y)) - f(x_0 + iy_0)}{i \Delta y}$$



$$= \lim_{\Delta y \rightarrow 0} \frac{u(x_0, y_0 + \Delta y) - u(x_0, y_0)}{i \Delta y} + i \frac{v(x_0, y_0 + \Delta y) - v(x_0, y_0)}{i \Delta y}$$

$$= -i u_y(x_0, y_0) + v_y(x_0, y_0)$$

Thus part (1) is proved.

ie, $f'(z) = v_y(x_0, y_0) - i u_y(x_0, y_0)$

In particular, $u_x(x_0, y_0) = v_y(x_0, y_0)$ &

$$u_y(x_0, y_0) = -v_x(x_0, y_0)$$

part 2 is proved,

On the other hand, if u_x, u_y, v_x, v_y exist at (x_0, y_0) & satisfy (19) CR eqns, it is not necessary that $f(z)$ is differentiable at $z_0 = x_0 + iy_0$.

Eg.
$$f(z) = \begin{cases} \frac{(\bar{z})^2}{z} & \text{if } z \neq 0 \\ 0 & \text{when } z = 0 \end{cases}$$

$$f(z) = \frac{(x-iy)^2}{x+iy} = \frac{(x-iy)^3}{x^2+y^2}$$

$$\Rightarrow f(z) = \frac{1}{|z|^2} \left[\underbrace{x^3 - 3xy^2}_{u(x,y)} + i \underbrace{(y^3 - 3xy^2)}_{v(x,y)} \right]$$

$$\text{Let } u(x,y) = \begin{cases} \frac{x^3 - 3xy^2}{x^2+y^2} & (x,y) \neq (0,0) \\ 0 & \text{o/w} \end{cases} \quad v(x,y) = \begin{cases} \frac{y^3 - 3xy^2}{x^2+y^2} & (x,y) \neq (0,0) \\ 0 & \text{o/w} \end{cases}$$

a) ~~the~~ Verify that CR eqns are satisfied at $(0,0)$.

However, $f'(0)$ doesn't exist.

$$u_x(0,0) = \lim_{\substack{h \rightarrow 0 \\ h \neq 0}} \frac{u(h,0) - u(0,0)}{h} = \lim_{h \rightarrow 0} \frac{\frac{h^3}{h^2} - 0}{h} = 1$$

$$u_y(0,0) = \lim_{k \rightarrow 0} \frac{u(0,k) - 0}{k} = \lim_{k \rightarrow 0} 0 = 0$$

$$v_x(0,0) = \lim_{h \rightarrow 0} \frac{v(h,0)}{h} = \lim_{h \rightarrow 0} 0 = 0$$

$$v_y(0,0) = \lim_{k \rightarrow 0} \frac{v(0,k)}{k} = \lim_{k \rightarrow 0} \frac{\frac{k^3}{k^2} \times k}{k} = 1$$

ie, $u_x = v_y$ & $u_y = -v_x$ at $(0,0)$.

b) Show that ~~$f(z)$~~ & $f'(z_0)$ doesn't exist. (Exercise)

~~By taking~~ partial derivatives, we can see that exist for u & v , for eg.

$$u_z(x,y) = \begin{cases} \frac{x^4 + 6x^2y^2 - 3y^4}{(x^2+y^2)^2} & (x,y) \neq (0,0) \\ 1 & (x,y) = (0,0) \end{cases}$$

Hence,

$$\lim_{h \rightarrow 0} u_x(h, 0) = \lim_{h \rightarrow 0} \frac{h^4}{h^4} = 1 \text{ but}$$

$$\lim_{k \rightarrow 0} u_x(0, k) = \lim_{k \rightarrow 0} \frac{-3k^4}{k^4} = -3$$

That is, $\therefore u_x$ is not continuous at $(0, 0)$.

~~Since~~ Hence, sufficient conditions for differentiability are not met. To prove that f is not differentiable, (see 20 Q9) notice that

$$\lim_{\Delta z \rightarrow 0} \frac{f(\Delta z) - f(0)}{\Delta z} \text{ doesn't exist as}$$

along $y=x$ line, say $(\Delta x, \Delta x)$

$$\lim_{(\Delta x, \Delta x) \rightarrow (0,0)} \frac{(\Delta x)^2 / \Delta x}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{f(\Delta z) - f(0)}{\Delta z} = \lim_{\Delta z \rightarrow 0} \frac{(\Delta z)^2}{(\Delta z)^2}$$

Along $y=x$ line: $\lim_{\Delta x \rightarrow 0} \frac{(\Delta x + i\Delta x)^2}{(\Delta x + i\Delta x)^2} = \lim_{\Delta x \rightarrow 0} \frac{(1-i)^2}{(1+i)^2} = -1$

but $\lim_{\Delta x \rightarrow 0} \frac{f(\Delta x, 0) - 0}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{(\Delta x)^2}{(\Delta x)^2} = 1$

(along x axis)

Pr 2) Show that the complex fn $f(z) = e^z$ is differentiable everywhere on $\mathbb{C}$.

$$f(z) = e^x \cos y + i e^x \sin y = u(x, y) + i v(x, y)$$

$$u_x = e^x \cos y \quad v_y = e^x \cos y$$

$$u_y = -e^x \sin y \quad v_x = e^x \sin y$$

Clearly, u_x, u_y, v_x, v_y exists everywhere on $\mathbb{R}^2$ & are continuous on $\mathbb{R}^2$. Since they satisfy CR eqns, $f(z)$ is differentiable everywhere on $\mathbb{C}$.

Sufficient condn. for differentiability

Theorem : Let $f(z)$ be defined throughout some ϵ -nbhd of a point $z_0 = x_0 + iy_0$. Suppose that

- u_x, u_y, v_x, v_y exist everywhere on the neighbourhood,
- u_x, u_y, v_x, v_y are continuous at (x_0, y_0)
- u & v satisfy the CR equations at (x_0, y_0)

$$u_x = v_y, \quad u_y = -v_x,$$

then $f'(z_0)$ exists & *

$$f'(z_0) = u_x + iv_x \big|_{(x_0, y_0)}.$$

~~Verify that $u(x,y) = \begin{cases} \frac{x^3 - 3xy^2}{x^2 + y^2} & (x,y) \neq (0,0) \\ 0 & (x,y) = (0,0) \end{cases}$~~

~~$v(x,y) = \begin{cases} \frac{y^3 - 3x^2y}{x^2 + y^2} & (x,y) \neq (0,0) \\ 0 & (x,y) = (0,0) \end{cases}$~~

~~doesn't satisfy CR eqns at $(x,y) \neq (0,0)$.~~

$$u_x(x,y) = \frac{(x^2 + y^2)(3x^2 - 3y^2) - (x^3 - 3xy^2)(2x)}{(x^2 + y^2)^2} = \frac{3(y^4 - x^4) - 2x^4 + 6x^2y^2}{(x^2 + y^2)^2}$$

$$= (x^4 + 6x^2y^2 - 3y^4) / (x^2 + y^2)^2$$

$$v_y(x,y) = \frac{(x^2 + y^2)(3y^2 - 3x^2) - (y^3 - 3x^2y)(2y)}{(x^2 + y^2)^2} = \frac{3(y^4 - x^4) - 2y^4 + 6x^2y^2}{(x^2 + y^2)^2}$$

$$= (y^4 + 6x^2y^2 - 3x^4) / (x^2 + y^2)^2$$

Thus, ~~$u_x \neq v_y$~~ if $u_x(x,y) \neq v_y(x,y)$ unless $x = \pm y$.

Show that a) $f(z) = 3x + y + i(3y - x)$ & b) $f(z) = \sin x \cosh y + i \cos x \sinh y$ are differentiable everywhere on $\mathbb{C}$.

- Let $f: D \rightarrow \mathbb{C}$ where D is an open connected set (domain) (23)
- 1) A function f of the complex variable z is analytic at a point z_0 if it has a derivative at each point in some neighbourhood of z_0 .

$$B(z, \epsilon) = \{ z \in \mathbb{C} \mid |z - z_0| < \epsilon \}$$

By defn, f is ~~not~~ analytic at $z_0 \Rightarrow f$ is analytic at each pt $z \in B(z, \epsilon)$.

- 2) A function f is analytic in an open set if it has a derivative everywhere in that set.

Note: If $S \subseteq \mathbb{C}$ is not open, we say f is analytic on S if f is analytic in an open set containing S .

- a) $f(z) = \frac{1}{z}$ is analytic ~~on $\mathbb{C} \setminus \{0\}$~~ at every point of $\mathbb{C}$ except 0.

- b) $f(z) = |z|^2$ is NOT analytic at any point.

Recall it is differentiable at $z_0 = 0$, but not on any cont neighbourhood containing 0.

- 3) A ~~entire~~ function is said to be entire if it is analytic at every point of $\mathbb{C}$.

Eg a) Polynomials $p(z)$ as derivative exists for them everywhere on $\mathbb{C}$.

- 4) ~~Singular point or singularity of f~~

~~If a function fails to be analytic at a point z_0 but is analytic at some~~

- 4) A necessary condition for analyticity of f in a domain D is satisfaction of CR eqns. ~~Sufficient~~ Sufficient conditions for analyticity in D of f in D is given in prev. thm.

- 5) ^{If} Two functions are analytic in D , then their

sum, product, quotient are respectively analytic in D provided $g(z) \neq 0 \forall z \in D$.

- 6) Composition of two analytic fns is analytic.

	function	domain	holo.	derivative
1)	$f(z) = z$	$\mathbb{C}$	✓	1
2)	$\frac{1}{z}$	$\mathbb{C} \setminus \{0\}$ Any S open subset of $\mathbb{C}$ of $\mathbb{C}$	✓	$-\frac{1}{z^2}$
3)	$\bar{z}$	$\mathbb{C}$	x	

(22)
(24)

Pr 1) Show that $f(z) = \frac{z^3 + 4}{(z^2 - 3)(z^2 + 1)}$ is analytic throughout the

z plane except at $\pm\sqrt{3}, \pm i$.

Derivative of f exists everywhere on $\mathbb{C} \setminus \{\pm\sqrt{3}, \pm i\}$ & hence is analytic here.

Pr 2) Show that $f(z) = \cosh x \cos y + i \sinh x \sin y$ is analytic on $\mathbb{C}$, i.e. entire

$$u(x, y) = \cosh x \cos y \quad \text{Recall} \quad \cosh x = \frac{e^x + e^{-x}}{2}, \quad x \in \mathbb{R}$$

$$v(x, y) = \sinh x \sin y \quad \sinh x = \frac{e^x - e^{-x}}{2}, \quad x \in \mathbb{R}$$

$$\text{Verify} \quad \frac{d}{dx} \cosh x = \sinh x$$

$$\frac{d}{dx} \sinh x = \cosh x$$

u_x, u_y, v_x, v_y clearly exist everywhere on $\mathbb{C} \cong \mathbb{R}^2$ & are continuous.

$$u_x = \sinh x \cos y$$

$$v_y = \sinh x \cos y$$

$$u_y = -\cosh x \sin y$$

$$v_x = \cosh x \sin y$$

$$\Rightarrow u_x = v_y \quad \& \quad u_y = -v_x.$$

Pr 1) Suppose that a function $f(z)$ & its conjugate $\overline{f(z)}$ are both analytic on a given domain D , then $f(z)$ must be constant throughout D . ①

Ans:

$$f(z) = u(x,y) + iv(x,y) \Rightarrow u_x(x,y) = v_y(x,y) \text{ on } D$$

$$u_y = -v_x$$

$$\text{Now, } \overline{f(z)} = u(x,y) - iv(x,y) \Rightarrow u_x(x,y) = -v_y(x,y)$$

$$\& u_y(x,y) = v_x(x,y) \text{ on } D$$

$$\text{ie, } u_x = v_y = -v_y \text{ on } D \quad \& \quad u_y = -v_x = v_x \text{ on } D$$

$$\Rightarrow u_y = v_x = 0 \text{ on } D$$

$$\Rightarrow v = f_1(x) \text{ on } D \quad (\text{as } v_y = 0)$$

$$\& v = f_2(y) \text{ on } D$$

$$\therefore v = c_2 \text{ on } D$$

$$\text{Similarly, } u = c_1 \text{ on } D \Rightarrow \underline{f(z) = c_1 + ic_2 = c} \quad \forall z \in D$$

~~Ex 1) s.t if $f(z)$ is diff at z , then $f(z)$ is cte at z .~~

Pr 2) If f is analytic on a domain D , and $|f(z)|$ is constant throughout D , then show that $f(z)$ must be a constant on D .

Ans: - Let $|f(z)| = c$.

$$\Rightarrow |f(z)|^2 = u^2 + v^2 = c^2. \quad \text{Either } c = 0 \text{ or } c \neq 0.$$

Case 1 $c \neq 0 \Rightarrow \overline{f(z)} = \frac{c^2}{f(z)}$. Since $|f(z)| \neq 0$ throughout D , $\overline{f(z)}$

is also analytic on D . Thus, by prev. problem, $f(z) = c'$ on D .

Case 2 $c = 0 \Rightarrow |f(z)| = 0 \text{ on } D \Rightarrow f(z) = 0 \text{ on } D$

Pr 3) Show Pr 2) ^{directly} using CR eqns without using Pr 1)

Let $|f(z)|^2 = u^2 + v^2 = c^2$ on D

(2)

$\Rightarrow 2uu_x + 2vv_x = 0$ & $2uu_y + 2vv_y = 0$

$\Rightarrow uv_y + vv_x = 0$ & $-uv_x + vv_y = 0 \rightarrow (2)$

① xv & ② $xu = uvv_y + v^2v_x + u^2v_x - uvv_y = 0$

$\Rightarrow (u^2 + v^2)v_x = 0$ on D .

But $u^2 + v^2 = c^2$. If $c = 0$, then $u(x,y) = 0$ & $v(x,y) = 0$ on D
 $\Rightarrow f(z) = 0$ on D .

If $u^2 + v^2 = c^2 \neq 0$ on D , then $v_x(x,y) = 0$ on D

$\Rightarrow v(x,y) = f_1(y)$ on D

~~iii³, ① xu + ② $xv = u^2v_y + v^2v_y = 0$~~

~~$\Rightarrow (u^2 + v^2)v_y = 0$~~

~~$\Rightarrow v(x,y) = f_2(x)$ on D~~

If so, by CR

$u_y(x,y)$ on D

$\Rightarrow u(x,y) = f(x)$ on D

&

ie, $v(x,y) = c_1$ on D

iii⁴, ① xu + ② $xv = u^2v_y + v^2v_y = 0$

$\Rightarrow (u^2 + v^2)v_y = 0$

$\Rightarrow v(x,y) = f_2(x)$ on D

ie, $v(x,y) = f(y) = f_2(x)$ on D

$\Rightarrow v(x,y) = c_2$ on D

iii⁴, ~~it can be~~ it can be also shown that

$u(x,y) = c_1$ on D

$\therefore f(z) = c_1 + ic_2$ on D .

④ Q. If $f(z) = z^2$, then what is the value of $u_{xx} + u_{yy}$ & $v_{xx} + v_{yy}$

$u(x,y) = x^2 - y^2$

$v(x,y) = 2xy$

$u_y = -2y$

$u_x = 2x$

$v_x = 2y$

$v_y = 2x$

$u_{yy} = -2$

$u_{xx} = 2$

$v_{xx} = 0$

$v_{yy} = 0$

$\Rightarrow u_{xx} + u_{yy} = 0$

$\Rightarrow v_{xx} + v_{yy} = 0$

Harmonic Functions

(3)

A real valued function $h: \mathbb{R}^2 \rightarrow \mathbb{R}$ is said to be harmonic in a given domain D of the xy plane if

i) it has continuous partial derivatives of 1st & 2nd order &

ii) it satisfies the Laplace eqn $h_{xx} + h_{yy} = 0$ on D

$$\nabla^2 h(x, y) = 0 \text{ on } D.$$

Lee 21 :-

Pr 2 If a function $f(z) = u(x, y) + iv(x, y)$ is analytic in a domain D , then its component functions u & v are harmonic in D .

For this, we assume the following facts:

If $f(z)$ is analytic at z_0 , then all its n^{th} derivatives, $f^{(n)}(z)$ exists at z_0 .

This will be further explored

~~Its proof will be taken~~ while developing Cauchy's integral formula.

Ans :-

Recall that if $f(z) = u(x, y) + iv(x, y)$, then

$$f'(z) = u_x + iv_x = v_y - iu_y$$

Since $f''(z_0)$ exists at z_0 , $f'(z)$ is cts at z_0 .

$\therefore u_x$ & v_y (also v_x & u_y) are cts at (x_0, y_0)

$$\begin{aligned} \text{Further, } f''(z_0) &= \left. \frac{\partial}{\partial x}(u_x) + i \frac{\partial}{\partial x}(v_x) \right| \frac{\partial}{\partial x}(v_y) + i \frac{\partial}{\partial x}(-u_y) \\ &= u_{xx} + iv_{xx} \quad \quad = v_{yx} - iu_{yx} \end{aligned}$$

$$\begin{aligned} \text{Also, } f''(z_0) &= \left. \frac{\partial}{\partial y}(-u_y) - i \frac{\partial}{\partial y}(v_y) \right| \frac{\partial}{\partial y} \\ &= -u_{yy} - iv_{yy} \end{aligned}$$

$$\text{i.e., } u_{xx} + u_{yy} = 0 \text{ on } D$$

$$\& v_{xx} + v_{yy} = 0 \text{ on } D$$

Further, since $f^{(3)}(z_0)$ exists, $f''(z_0) = u_{xx} + i v_{xx}$ is

(4)

$f^{(3)}(z_0)$ is continuous at z_0 & hence u_{xx} & v_{xx} are also

at z_0 & hence, $u_{xy} = u_{yx}$ & $v_{xy} = v_{yx}$ also. we get $(u_x)_x = (v_y)_x = v_{yx}$
(iii) $u_{xy} = u_{yx}$ too $\Rightarrow (-u_y)_y = (v_x)_y = v_{xy}$

Corollary If $f(z) = u(x, y) + i v(x, y)$ is analytic at $z_0 = (x_0, y_0)$, then u & v have continuous partial derivatives of all orders at that point.

Theorem A function $f(z) = u + i v$ is analytic in a domain D iff v is a harmonic conjugate of u .

Defn If two given functions u & v are harmonic in a domain D & their first order partial derivatives satisfy C.R. eqns throughout D then v is said to be a harmonic conjugate of u .

Proof $\Rightarrow f(z)$ is analytic in D

$$\Rightarrow u_x = v_y \text{ \& } u_y = -v_x \text{ on } D$$

$$\text{ \& by pr 3), } u \text{ \& } v \text{ satisfy } u_{xx} + u_{yy} = 0 \text{ on } D$$

$\Rightarrow v$ is a harmonic conjugate of u

$\Leftarrow$ By defn of harmonic fns, $u_x, u_y, v_x, v_y, u_{xx}, u_{yy}$ (\& hence v_{yx} \& v_{xy}), v_{xx} \& v_{yy} (\& hence u_{yx} \& u_{xy}) exist \& are cont ~~exist~~ on D . further $u_x = v_y$ \& $u_y = -v_x$ on D . If all first order partial derivatives are continuous in D , we ~~are~~ have ~~by~~ that $f(z)$ is analytic on D . However, since $f'(z)$ is continuous on D , ~~its~~ its real part u_x \& img. part v_x are also cont ~~at~~ on D \& hence, $f(z)$ is analytic on D .

Finding harmonic conjugate v of u

$$\text{Let } u(x, y) = y^3 - 3x^2y$$

Check that it is harmonic on $\mathbb{C}$.

Let v be a harmonic function conjugate of u . Then, automatically v is harmonic on D & ~~$v \in C^1(D)$~~ satisfies CR eqns.

$$\therefore v_x = -u_y = -3y^2 + 3x^2 \quad - (1)$$

$$\& v_y = u_x = -6xy \quad - (2)$$

Then, $v = -3y^2x + x^3 + c_1(y)$ from (1)
 $= -3xy^2 + c_2(x)$ from (2)

$$\Rightarrow v(x, y) = x^3 - 3xy^2 + C$$

& hence, $f(z) = u + iv$ is analytic on D , i.e. an entire function.

$$= i(y^3 - 3x^2y + i(x^3 - 3xy^2) + C)$$

This can be seen to be equal to $f(z) = i(z^3 + C)$ by verification.

To guess: put $y=0$. Then $f(x) = i(x^3 + C)$

FACT :- Any harmonic function on an open ball $B(a, r)$ has a harmonic conjugate in $B(a, r)$. (More generally, in simply connected domains)

2) Harmonic ~~find~~ conjugate of a harmonic function $u(x, y)$, when exists, is unique except for an additive constant. (Ex 2 sec 26)

Elementary functions

1) Exponential function

Recall $e^z = e^x e^{iy}$ where $e^{iy} = \cos y + i \sin y$ (y radians)

Note that ~~$e^{\frac{1}{n}}$ was defined as all $(n$ many) distinct n^{th} roots of e . This was a set. However, when $z = \frac{1}{n} \ln(3)$, we write e^z where $x = \frac{1}{n}$, we consider $\sqrt[n]{e}$, the n^{th} root of e .~~

~~This~~ ~~expose~~ The complex exponential function is an extension of the real exponential function e^x .